

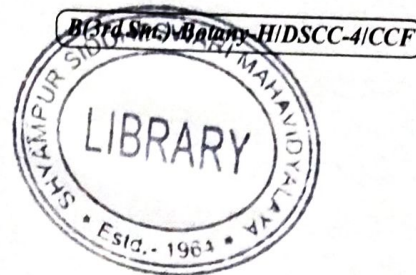
2024

BOTANY — HONOURS

Paper : DSCC-4

(Plant Anatomy and Embryology)

Full Marks : 75



The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

1. Answer **any six** questions from the following :

2×6

- (a) Define Plastochron.
- (b) How Phelloderm is formed?
- (c) Define Apospory and Apogamy.
- (d) What is 'heartwood'?
- (e) Distinguish between leaf trace and leaf gap.
- (f) Distinguish between Apoplast and Symplast.
- (g) Define Apomixis.
- (h) What is Microgametogenesis?
- (i) Distinguish between fascicular and interfascicular cambium.

2. Answer **any three** questions from the following :

5×3

- (a) Discuss the types of stomata according to Stebbins and Khush (1961).
- (b) Write a note on the secondary growth found in monocot stem.
- (c) Give an overview of the ultrastructure of the cell wall of vascular plants.
- (d) Write a note on adaptive anatomical features of xerophytes.
- (e) Briefly describe three types of endosperm developments found in angiosperm.

3. Answer **any four** questions from the following :

- (a) Write a note on mechanical tissues found in different plant organs. Explain with suitable diagrams the principles governing their distribution in plants. 4+8
- (b) Define stele. Describe different types of stele with examples and diagrams. 1+6+2+3
- (c) Discuss the secondary growth in the stems of *Bignonia* and *Tecoma* with diagrams. Comment on their adaptive advantages. 4+4+4

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- (d) Diagrammatically represent the Korper-Kappe theory in relation to root apex organization. Write the adaptive anatomical features of halophytes. 8+4
- (e) What do you mean by 'Dendrochronology'? Discuss, in brief, the different applications of plant anatomy in Systematics and Forensics. 2+5+5
- (f) With suitable diagram describe the embryogenesis in *Capsella*. Describe, in brief, the tetrasporic type of embryo sac development. 8+4
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